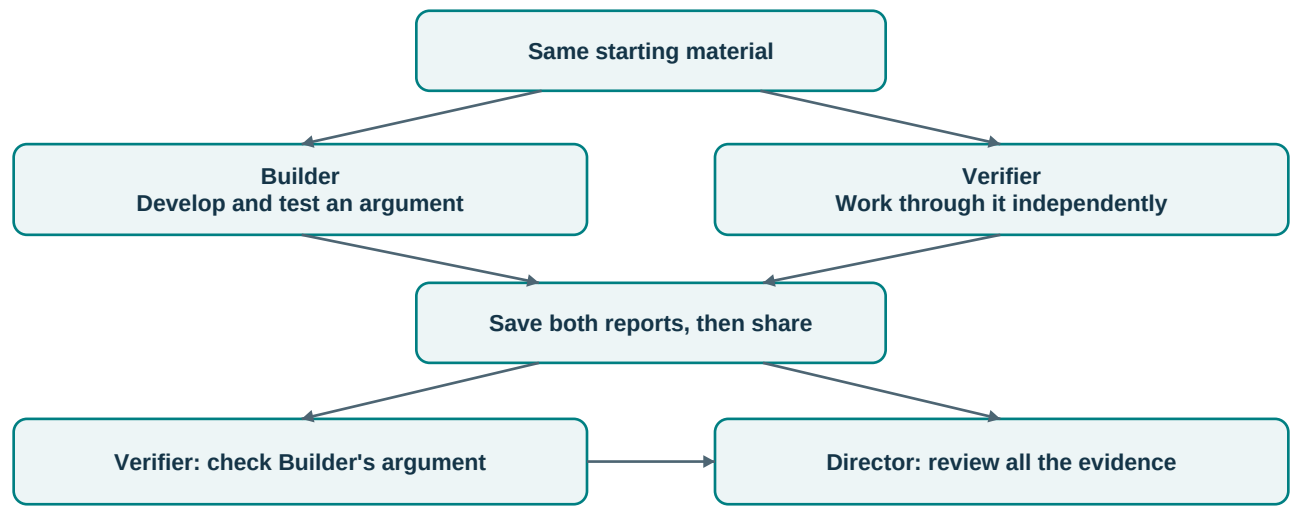


ARCHITECTURE NOTE

Theory Lab

Theory Lab gives three agents different jobs. Builder develops an argument. Verifier works through the same problem independently, then checks Builder's work. Director reviews the evidence and decides what belongs in the shared research record. Agreement alone is not enough: a claim needs a clear statement, explicit assumptions, and evidence that supports it.

This guide requires Builder and Verifier to finish and save both reports before either report is shared. Later corrections belong in separate notes, so readers can still see what each agent found independently.



Update the research record. The human researcher decides on major changes.

Responsibilities

Component	Responsibility	Limits
Builder	State claims precisely, develop arguments, test assumptions, and keep useful failed attempts.	Writes its own report and recommends changes to the shared record.
Verifier	Work independently, then check Builder's reasoning, assumptions, counterexamples, and use of sources.	Saves its independent report before reading Builder's work; cannot approve its own claim.
Director	Compare the claims, weigh the evidence, and record a decision.	Must follow the review requirements and refer substantial changes to the human.
Controller or operator	Save the starting material and reports, control access, check record structure, and track resource use.	These controls do not establish mathematical correctness.
Human	Set the objective, decide substantial changes, review progress, and choose when to stop or continue.	Agents may recommend these decisions but cannot make them for the human.

OPERATING PROTOCOL

One round

1. **Prepare the starting material.** Save the objective, statements to examine, allowed sources, current research record, and resource limits. Give this copy a unique reference so everyone can identify what was supplied.
2. **Work independently.** Give Builder and Verifier the same starting material in separate contexts. Neither reads the other's work. Each explains its assumptions, supporting claims, evidence, and remaining gaps.
3. **Finish and save both reports before sharing.** Keep the saved copies unchanged and record their content hashes. A hash helps identify a file; access controls must keep the reports private until this step is complete.
4. **Check Builder's work.** Verifier reads Builder's saved report and writes a separate critical review. Find invalid steps or mismatched assumptions, check cited theorems, and test counterexamples against every stated assumption.
5. **Review the evidence.** Director reads both reports and the critical review. Check whether the agents considered the same statement, assess each objection, and explain the decision. One valid counterexample can defeat a claim even when several agents agree with it.
6. **Update the shared research record.** Check the records and references, then apply only changes that meet the review requirements. Refer substantial changes in objective, scope, or assumptions to the human. Keep earlier statements, refutations, and reports available.
7. **Review progress.** Record which claims held up, which failed, and what remains unresolved. Use a saved copy of the updated record for any later round. Report to the human after every three completed rounds, or sooner if a substantial decision blocks progress.

Research records

The shared research record contains definitions, assumptions, and claims. Keep it separate from reports, which retain each round's reasoning and decisions, and from generated indexes, which help readers find records.

Give each claim a stable identifier and a version number. Record the exact statement, definitions, assumptions, claims it relies on, evidence, status, unresolved objections, and the decision behind its status. When using an outside theorem, cite the source and explain how its assumptions match the present use.

Distinguish a written proof, a cited theorem, a finite computation, and a conjecture. For a computation, record the domain, range, procedure, random seed when relevant, and what the result shows. A finite check supports only what it checked; a broader conclusion needs a separate argument.

Changing an assumption, quantifier, domain, or conclusion creates a new statement version and requires another review. Keep a refuted statement unchanged. Give a repaired statement a new identifier linked to the earlier one. If a supporting claim changes, review the claims that depend on it.

KEEPING TRACK OF THE WORK

How claims are reviewed

State	Meaning
REAUDIT_PENDING	A claim from another source is waiting for review.
OPEN	A question is clear, but a complete claim has not yet been stated.
CONJECTURED	The claim and everything it assumes or relies on are stated explicitly.
COMPUTATIONALLY_SUPPORTED	Reproducible finite checks support the claim within the checked range.
PROVED_PENDING_AUDIT	A complete written proof is available, but independent review is unfinished.
ESTABLISHED	The required independent checks are complete and Director has accepted the claim.
REFUTED	A valid counterexample or contradiction defeats the stated claim.

Accepting a claim as ESTABLISHED requires a saved statement and content hash, explicit definitions and assumptions, a complete written proof, an independent review of that same statement, and accepted supporting claims. It also requires applicable attempts to find counterexamples, careful checks of outside theorems, resolution of any substantial decision affecting the claim, and Director's written decision. No agent may approve its own result.

The status policy lists the allowed changes between states. ESTABLISHED means the claim passed this review process. It does not mean the proof was checked by a proof assistant or that the reviewers cannot be wrong.

Running the workflow

- Control what each role can read and edit. Different prompts or folders alone do not keep the agents' working contexts separate.
- Keep both saved reports unchanged, and record when they are shared. Put later corrections in separate notes.
- Give each role only its assignment, allowed sources, its own work, and reports shared at the proper stage. Keep credentials and operator-only material out.
- Set resource limits, track actual use, and stop at those limits. Keep the work completed so far and note what remains unresolved.
- Check records and references before applying changes. Preserve earlier versions and the reasons for decisions.
- Treat documents, citations, and tool outputs as material to examine. They cannot grant permissions or change a role's instructions.

These are rules for running the workflow. This repository provides instructions, not software that enforces them; a working setup must implement and check the controls described here.